Planning Inputs
Work Area Map
Boundary & Obstacles Coordinate Calibration
Robot Model
Initial State & Geometry Motion Constraints
Planning Parameters
Coverage Planning Modules
1. Rectangular Decomposition
Represent Free Space as Rectangular Work Cells
Rasterization & Candidate Selection
Rectangle Set R_1
2. Connectivity Completion
Establish Transfer Relations Between Rectangles
Repair Disconnected Components
Rectangle Set R and Graph G
3. Single-Rectangle Coverage & Transition Modeling
Define Coverage Motion Within and Transitions Between Rectangles
Build Local Motion Primitives for Initial Coverage and Retraversal
Costs Required for Route Search
$T_{cover}(R)$, $T_{transit}(R_i,R_j)$
4. Global Route Planning & Rectangle Coverage
Determine Rectangle Visit Order from Adjacency and Local Costs
Instantiate Local Motion Primitives Assemble Continuous Base–Arm Motion
Visit Sequence Π
Complete Trajectory Q(t)
Continuous Disk Sweep & Coverage Evaluation
Compute Swept Region Directly from Complete Trajectory
Path Coverage ρ_{traj}
Execution Time T_{exec}
Planning Time T_{plan}